Mining Health, Safety and Equipment Monitoring System

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October 2026

Abstract

Mining operations require effective monitoring of worker safety, incidents, equipment condition and maintenance activities. This project developed a Streamlit-based Mining Health, Safety and Equipment Monitoring System to demonstrate how these activities can be supported through a single application. The system includes role-based access, worker safety records, incident management, equipment monitoring, maintenance tracking, risk assessment and reporting with downloadable CSV data. Initial functional testing confirmed that the main pages load and that the equipment,maintenance, risk assessment and reporting features tested behaved as expected. The application is an academic prototype and has not been certified for operational mine use.

1 Introduction

Health and safety are major priorities in the mining industry. Hazards, equipment failures and incomplete incident records can increase operational risk. Digital monitoring tools can help organise relevant information and support timely review by authorised personnel. The aim of this project was to develop an interactive application that brings together selected mining safety and equipment monitoring functions. Its objectives were to record worker safety information, manage incidents, display equipment condition indicators, track maintenance, assess risk and provide reports.

2 Background

Mining safety management involves identifying hazards, assessing risk, recording incidents and ensuring that workers follow appropriate safety procedures. Equipment condition monitoring also supports maintenance planning by helping personnel identify readings or conditions that may require attention. A centralised information system can improve accessibility and make routine review more convenient. However, software outputs must be interpreted alongside

site procedures, professional judgement and applicable safety requirements. A student prototype cannot replace certified monitoring equipment or formal mine safety controls.

3 Methodology

The application was implemented in Python using Streamlit to provide a graphical user interface. The project was organised around the following functional modules:

- **Role-based access:** separate permissions for Administrator, Safety Officer, Mining Engineer, Maintenance Engineer and Manager.
- **Worker safety:** records for worker details, training, personal protective equipment (PPE), fatigue and safety observations.
- **Incident management:** a page for recording and reviewing safety incidents.
- **Equipment monitoring:** equipment records, temperature and vibration readings, and alert statuses.
- **Maintenance:** maintenance records and task status information.
- **Risk assessment:** likelihood and consequence inputs used to display a risk score or level.
- **Reporting:** analytical summaries and CSV downloads for further review.

Testing was performed manually through the application interface. The equipment monitoring, maintenance, risk assessment and reporting pages were checked, along with Administrator access to user management. Testing focused on whether pages loaded and whether the described outputs or actions were available.

4 Results and Discussion

The pages tested opened successfully. Equipment Monitoring displayed equipment records, temperature and vibration readings, and alert statuses. Maintenance displayed maintenance records. Risk Assessment accepted likelihood and consequence selections and displayed a risk score or level. Reporting Analysis displayed analysis and successfully downloaded CSV data. The Administrator account could access the Manage Users page. These results demonstrate that the principal tested workflows were accessible in the running application. They do not establish that the application is secure against all threats, that every possible input has been tested, or that its alerts are suitable for real-world operational decisions. Further systematic testing is needed, including validation of calculations, invalid inputs, data persistence, security and permission enforcement for every role.

4.1 Development process and architecture

The work followed the sequence in the brief: define the problem, design the algorithms, draw flowcharts, write pseudocode, code the functions, then build and test the interface. The code is split into three files so that each part can be tested separately

4.2 Dataset

No real mine data were available, so `generate_data.py` creates a synthetic dataset with a fixed random seed so results can be repeated: 60 workers, 120 incidents over twelve months, 22 pieces of equipment (haul trucks, loaders, excavators, drilling machines, bulldozers, crushers, conveyors and a scraper winch) and five user accounts. Passwords are stored as SHA-256 hashes. The incident data give the night shift slightly more incidents so that the day-and-night comparison has something to show. The synthetic readings are not meant to represent a real fleet.

5 Conclusion and Recommendations

The project produced a working Streamlit application that demonstrates an integrated approach to selected mining safety and equipment monitoring tasks. Initial manual tests confirmed the main functions described in the results section. Future improvements should include broader automated testing, stronger authentication and audit logging, validation against relevant safety procedures, improved data backup and persistence, and testing with realistic datasets. Before any operational use, the system would require expert review, appropriate cybersecurity controls and formal validation and certification where applicable.

References

[1] African Review (2026) *New reports outline roadmap for South African mining modernisation*. African Review of Business and Technology, 15 July. https://www.africanreview.com (Accessed: 21 August 2026).

[2] HSRC (2026) *South Africa's Mining Future Takes Centre Stage at Modernisation Day*. Human Sciences Research Council, 12 June. https://www.hsrc.ac.za (Accessed: 21 August 2026).

[3] PwC South Africa (2026) *Ten Insights into 4IR in South African Mining*. PwC Publications, 20 May. https://www.pwc.co.za (Accessed: 21 August 2026).

[4] Madahana, M. and Phala, T. (2026) *MINN2020A Project: Design, Development and Documentation of an African Critical Minerals Application*. School of Mining Engineering, University of the Witwatersrand, 18 August.

[5] Streamlit (n.d.) *Streamlit documentation*. https://streamlit.io (Accessed: 21 October 2026).

[6] pandas (n.d.) *pandas documentation*. https://pandas.pydata.org (Accessed: 21 October 2026).

[7] Plotly (n.d.) *Plotly Python graphing library*. https://plotly.com/python (Accessed: 21 October 2026).